Day 5: mathematical diagnostic

Name: _____ Date: _____

60 minutes total. No calculator, notes, internet or AI on the independent pass. Show your reasoning. If stuck, leave the useful work you have and move on. This is not a speed contest.

3 minutes setup · up to 30 minutes core · up to 20 minutes for two selected follow-ups · 7 minutes for your log. The tutor will supply follow-ups. Do not work through the whole reserve book.

If you finish a block early, check your work or continue to the next block with the tutor. Do not exceed the overall limit.

L1-3

A student says: “n^2 is even, so n is even, because squaring an even number gives an even number.” Is the claim true, and is that argument enough? Repair it.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

L2-3

Over the integers, compare: (A) for every x there exists y with $y>x$; (B) there exists y such that for every x, $y>x$. Which are true?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N1-3

Explain why a decimal integer is divisible by 11 iff its alternating digit sum is divisible by 11.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

L1-2

For integers n, assess both claims: if 4 divides n then 2 divides n; if 2 divides n then 4 divides n.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

L2-2

Negate precisely: “For every integer n there exists an integer m with m>n.” Is the original statement true?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N1-2

For which digit d is the decimal number 4d2 divisible by 9?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

L2-1

Negate: “For every real x, P(x) holds.” Negate: “There exists a real x for which P(x) holds.”

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N1-1

State the tests for divisibility by 3 and 9 for a nonnegative integer in decimal notation.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

Session log

Which question was hardest to start, and what did you try?

_____

Was any question familiar, hinted, interrupted, or affected by outside study? Record IDs and details.

_____

Follow-up IDs attempted: _____ / _____